Multilayer Perceptron for Predicting Yield

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1 Performance

1.1 Metrics

- **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values. A lower MSE indicates better model performance.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2 = 1.54 \times 10^7$$

- **Root Mean Squared Error (RMSE):** The square root of MSE, bringing the error back to the same units as the output variable (yield).

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} = 3926.4902$$

- **Coefficient of Determination (R²):** Represents the proportion of variance in the target variable that is predictable from the input features.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} = 0.9452$$

- **Pearson Correlation (r):** Measures the strength and direction of the linear relationship between predicted and actual values.

$$r_{\mathrm{Pearson}} = \frac{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)\left(\hat{y}_i - \bar{\hat{y}}\right)}{\sqrt{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2}\sqrt{\sum_{i=1}^{n}\left(\hat{y}_i - \bar{\hat{y}}\right)^2}} = 0.9681$$

Table 1: Summary of Important Parameters

Metric / Parameter	**Value**
Training years	2010–2019
Validation years	2020–2021
Target prediction year	2022
Number of unique countries	165
Number of unique crop items	102
Training samples	52,370
Validation samples	10,368
Prediction samples (2022)	5,190
Final Validation R^2	0.9452
Final Validation Pearson r	0.9681
Final Validation MSE	1.54×10^7
Final Training MSE	9.62×10^6
Model architecture	2 Hidden Layers (128, 64) + ReLU + BatchNorm
Optimizer	Adam
Learning Rate	0.001
Batch Size	64
Loss Function	Weighted MSE
Epochs	120
Decay Weight Formula	$w = e^{-\lambda \cdot (\text{target year}} - \text{year})$
Decay Rate λ	0.1

1.2 Training vs Validation

The goal was to predict crops yield values (in kilograms per hectare) with multilayer perceptron (MLP) with the weather data from the past year. On the validation set (years 2020–2021) the model obtained its best coefficient of determination (R^2) of 0.9452, MSE $= 1.54 \times 10^7$, and final Pearson correlation of 0.9681 between predicted and actual yields. Data from 2010–2019 was used for training and 2020–2021 was held out for validation thus the splitting was not randomly selected and it was based on the year to make sure we are forcasting based on a year and there is no data leakage.

As we can see in 1, both training (blue) and validation (olive) loss start relatively high (on the order of 10^9 in MSE, due to the yields being in the tens of thousands) at epoch 0. The loss changed fast in the first 10 epochs because the model quickly learns relationships in the data. The loss continues to decline and by epoch 116 (val-loss: $\approx 1.1 \times 10^8$) both training and validation losses have nearly leveled off and the major changes happend in first 20 epochs. The validation loss is higher than the training loss because the model is still trying to fit the data. In the final epoch the training MSE is lower than the validation MSE. The final validation MSE is 1.54×10^7 and train MSE is 9.62×10^6. The reason why we set the max epoch to 120 was because the validation loss curve began to plateau before epoch 120 and the performance improvement was low. The validation R^2 did not show gain in epochs after 120 and if we continued training after that, we would have let to overfitting.

The table 2 shows the data predicted for 2021 and we can see the model's performance. Predictions were back-transformed from log-space to the original yield scale before evaluation. The sample in 2 is not representative of the full validation set.

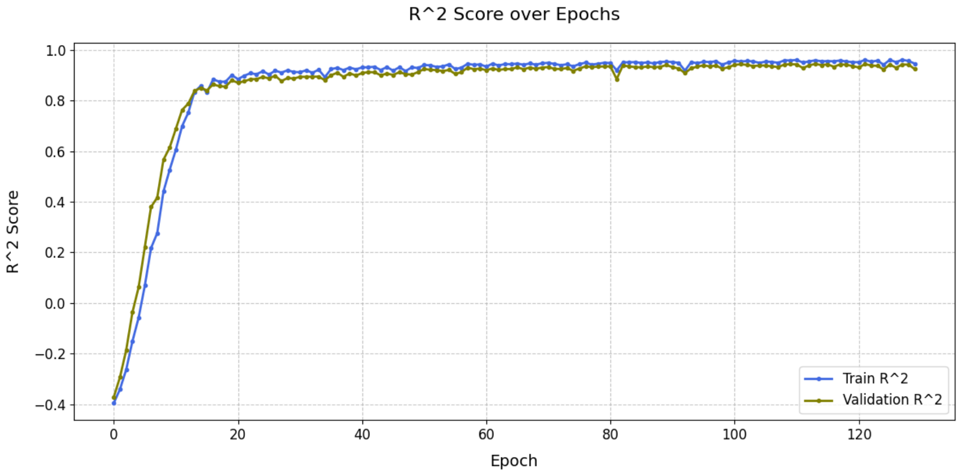

Figure 1: The mean R^2 during training epochs.

Country	Predicted	Actual	Residual	Item
Algeria	1997.4763	1731.3000	-266.1763	Triticale
Algeria	1829.4769	1717.0000	112.4769	Treenuts, Total
Angola	642.3450	643.5000	1.1550	Groundnuts, excluding shelled
Angola	4714.5840	4749.8000	35.2160	Other fruits, n.e.c.
Angola	744.1731	710.0000	-34.1731	Sunflower seed

Table 2: Predictions vs Actual for Year 2021

2 Model

This model is fully connected feed-forward neural network (MLP) consisting of an input layer, two hidden layers, and an output layer. The input layer is a vector of features which is passed through two hidden layers with 128 and 64 neurons, then a single output neuron will give us the results. Each hidden layer uses the ReLU activation function, and each is followed by a batch normalisation layer to stabilise learning. The output layer gives us a continuous numeric output. The model was trained on a dataset of 52,370 training rows and evaluated on a validation set of 10,368 rows which is for the years 2010–2019 and 2020–2021, respectively. The architecture is like below:

- Input layer: $\mathbf{x} \in \mathbb{R}^d$ (with d input features)
- Hidden layer 1: fully-connected linear mapping $\mathbb{R}^d \to \mathbb{R}^{128}$, BatchNorm on 128 features, then ReLU activation
- Hidden layer 2: fully-connected linear mapping $\mathbb{R}^{128} \to \mathbb{R}^{64}$, BatchNorm on 64 features, then ReLU activation
- Output layer: fully-connected linear mapping $\mathbb{R}^{64} \to \mathbb{R}^1$ (yield prediction)

This architecture is enough to capture non-linear interactions but not overly complex. The two layers of ReLU neurons are used to learn combinations of the input features. The batch normalisation after each hidden layer helps overcome internal covariate shift and lets us have a higher learning rates. No dropout layers were used and after numerous iterations it is found that dropout was not necessary for preventing overfitting and batch normalization prevents overfitting.

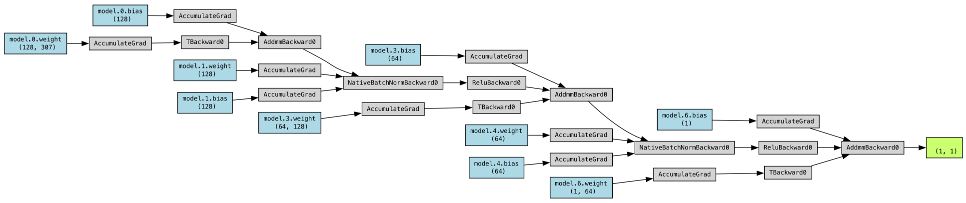

Figure 2: The architecture of the network with mentioned layers

2.1 Hyperparameters and Training

Adam optimizer is used in here with a learning rate of 0.001 because we had sparse gradients and this optimizer has the ability to self-adjust the learning rate for each parameter. The loss function was Mean Squared Error (MSE) good for a regression task that we want to minimise the squared difference between predicted and actual yields. Training was done in mini-batches of size 64 to balance speed and noise. The training process ran for up to 120 epochs but the model converged well before 116 epochs as we can see by loss plot ($R^2 = 0.9452$, val-loss $\approx 1.54 \times 10^7$) but the training was continued to epoch 120. The loss for each batch was computed and then weighted by a factor before backpropagation and this was done by multiplying the criterion output by a weight vector w_batch and taking the mean. This means more recent data points contributed more strongly to the weight updates than older data points.

2.2 Model Performance

After each epoch's weight updates the model was evaluated on the validation set to compute the validation loss and R^2. Although we did not had early stopping training when validation loss stops improving implemented in code, printing loss and R^2 let us stop training. Validation performance improved or remained stable during epochs and we can tell that the model was effectively trained. Training could be stopped early if the validation loss started to increase which is a sign of overfitting. Small spikes can happen in validation loss when the model adjusts weights that overfit the training data and temporarily perform worse on unseen data.

3 Features and Labels

Below is the complete list of input features used:

- Year: From 2010 to 2022
- Country: After merging all the datasets we got 165 countries. Year, Country, and Item are related and it is mentioned in Preprocess Section
- Crop Item: The crop for which yield is being predicted. By including the crop type, the model can differentiate yield levels and sensitivities among different crops.
- Soil Moisture - 4 Columns: Annual mean soil moisture at depth of 010 cm, 1040 cm, 40100 cm, and 100200 cm in the soil. These features are calculated from mean of monthly soil moisture data. Soil moisture shows water availability in soil.

- Soil Temperature - 4 depth, 3 column for each: Annual mean soil temperature, minimum soil temperature, and maximum soil temperature for 4 depth (010, 1040, 40100, 100200).
- Evaporation from Soil (ESoil): Annual mean soil evaporation rate (ESoil_tavg) which is moisture evaporated from the soil surface and monthly values were averaged into an annual mean.
- Transpiration (TVeg): Annual mean transpiration rate or the water loss through plant transpiration.
- Canopy Water Content (CanopInt): Annual mean canopy water interception which is calculated from monthly values.
- Total Water Storage (TWS): Annual mean terrestrial water storage anomaly showing the sum of all forms of water storage.
- Land Cover Percentages - 17 Columns: The fraction of land area occupied by different land cover classes. These were merged by location and then averaged per country.
- Precipitation Rainfall: Annual total rainfall. This was computed by summing monthly rainfall over the year after converting units which is mentioned in preprocessing section.
- Precipitation – Snowfall: Annual total Snowfall. This was computed by summing monthly Snowfall over the year after converting units which is mentioned in preprocessing section.
- Output Label (Target Variable): The target is to predict the crop yield for the following year. Yield is the amount of crop produced per unit area harvested. The features from year t are paired with the yield from year $t + 1$. This was done by decrementing the year in the yield data by 1 before merging with features.

4 Preprocessing

4.1 Handling Missing Values

Some of the weather datasets contained missing values for certain months and due to the high number of missing values (14946 rows) we could not remove it. Rather than dropping those rows and reducing data coverage targeted imputation was applied:

- Country latitude and longitude data had missing values or zero centroid radius for 7 rows and these were removed from the dataset.
- Total Water Storage (TWS) data in month 2 and month 3 had missing values for 14946 rows. These were imputed using weighted average between months 1 and 4. February's TWS data filled as two-thirds of January's value plus one-third of April's and vice versa for March.

- Soil Moisture (100–200 cm) data in the 8th month (August) had missing for 14946 rows. These were imputed by taking the average of month 7 (July) and month 9 (September).

Any remaining missing values after merging all features were dropped. After merging all weather data with land cover all rows with any null were dropped because we did not had any weather data for the points that were available in land cover dataset. The yield dataset itself did not have missing yields for the years and countries considered and we just filtered the Elements that were Yield. Linear insertion was chosen because the missing gaps were short and the weather of each month is similar to the adjacent months.

4.2 Unit Conversions and Adjustments

Precipitation (Rainfall and Snowfall) had kilograms per square meter per second as their unit and the numbers were small causing problems in further steps. These numbers were converted to millimeters of precipitation per month by multiplying them to seconds, minutes, hours, the number of days in each month and then dividing it by 1000 supposing that density of water is 1 kilogram per square meter. After conversion, the 12 months were summed to get total annual precipitation in mm

Soil Temperature was in Kelvin and it was converted it to degrees Celsius by subtracting 273.15 and again the scale of the number would have caused problems in further steps because temperatures in degrees Celsius are more interpretable than Kelvin values. The mean, min, and max were calculated and then the conversion was applied. Other variables were already in coherent units.

Land cover percentages: One column had a minor naming issue classh_4 instead of class_4 which was fixed by renaming but no unit conversion was needed since they are percentages.

Yield in the dataset was in Kilogram per Hectare. We applied a log transform $\log(1+x)$ to yield values and 1 is just to avoid $\log(0)$.

Converting precipitation to mm and temperature to celsius makes the features have magnitudes that are easier to handle and more interpretable. It avoids extremely small or large values that could create numerical difficulties.

4.3 Feature Engineering

A key preprocessing step is converting monthly data into annual features. For each climate variable with monthly data we computed annual summaries. Annual mean was chosen for most variables but for precipitation (rain and snow) annual sum were computed because total rainfall is more relevant than average rate. For soil temperature Minimum and maximum were also added next to mean. This could show temperature stress throuoghout the year which could affect overwintering or summer heat stress on crops. This not only simplifies the model's input space but also corresponds to the fact that yields are reported yearly and using annual summaries is logical and it would also reduce the number of input features avoiding overfitting. The code uses functions calc_mean, calc_temp, and calc_prec_sum to apply the mentioned approaches to data.

4.4 Data Merging

All weather data were merged on common keys: latitude, longitude, and year (mergedWeatherdf) and the land cover percentages were merged into this by lat-long-year as well. Then each lat-long point was assigned to a country by choose_country function which uses Haversine formula to find the distance between each centroid of countries from the points and see which point lands in which country (by keeping radius in mind). Then if we had one or multiple countries which contain the point (because countries are assumed as circles) we choose the country in which the point lies most centrally is selected. For the points that does not have any countries we calculate the radius and see which country's border is closer to the point. After adding Country to column to each point's data latitude and longitude were dropped.

The MLP expects a fixed vector of input features and because of that dataset is grouped by country and year and took the mean of all other columns. This compressed the data so we would have one row per country-year. The grouping is necessary because the yield data is country level and climate features should also be in the country level.

The yield data was merged with the country-year climate data but before merging we need to shift back by one year. A yield recorded for $Year = T$ is now linked to features with $Year = T - 1$ and the merge was an inner join to just keep the country-years that have both climate data and yield data. The result is combined_df and now we have three dataframes, Train which is [2010 to 2019], Test which is 2020 and 2021, and Predict which is 2022 from the weather data and yield (just to keep items) and then dropping value column.

The final shapes:

- Item unique: 102,
- Country unique: 165
- train shape: (52370, 43)
- val shape: (10368, 43)
- pred shape: (5190, 42)
- Unique Years train: [2010 2011 2012 2013 2014 2015 2016 2017 2018 2019]
- Unique Years val: [2020 2021]

4.5 Exponential Decay Weighting

For empasising more on recent data the decay weighting strategy was used by introducing a new feature year_diff defined which is the difference between the target year and the year of each passed data point. In this case the target year was 2022 and decay weight for each sample was calculated using $w = e^{-\lambda \cdot (\text{year\_diff})}$ and the decay factor was set to $\lambda = 0.1$. This formula helps us use weights close to 1 for recent years and smaller weights for older data.

4.6 Feature Scaling

All continuous features were standardised using a StandardScaler which subtracts the mean and divides by the standard deviation. This helps to show the features on the same scale. Then a ColumnTransformer was applied using StandardScaler and categorical features were handled in here with one-hot encoding to incorprate categorical variables into the model and simply ignore any unseen categories in future data. This transformation resulted in a sparse matrix consisting of scaled numeric features and binary one-hot encoded columns. The feature vector would expand considerably by this method but it allows the model to have separate weight parameters for each country and crop.

4.7 Train Test Split

The combined dataset was split into training and validation sets based on Year. All records with Year 2010–2019 were taken as the training set, and records with Year 2020–2021 as the validation set as df_train.csv and df_test.csv. df_predict was 2022 weather data which is came from isolating year 2022 in merged_weather_compress and then getting the yield items and merging it.